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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Linux OS and Scripting – 5_Lab

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23/09/2026



MANIPAL SCHOOL OF INFORMATION SCIENCES

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(A constituent unit of MAHE, Manipal)

Part A: Authentication Log

Exercise 1: To print

Write sed commands to:

Print the first log entry

Print the fifth log entry.

Print entries from lines 2 to 6.

Print the last entry.

Print every second log entry starting from line 1.

Print all entries containing Failed password.

Print all successful login entries containing Accepted password.

Print entries from the first failed login to the authentication failure.

```

haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
~/Documents/LoS/Lab/Lab5

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '1p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '5p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '2,6p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '2,6p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '5p' auth.log
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$

```

Exercise 2: Filtering Security Events

Write commands to:

Display failed login attempts.

Display successful login attempts.

Display entries associated with the root account.

Display entries containing Invalid user.

Display all events related to IP address 172.16.0.8.

Display all events that do not contain Accepted password.

Display all events except those generated by sudo.

Display only the entries that contain either Failed password or Invalid user using multiple `-e` commands.

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/root/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/Invalid user/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/172.16.0.8/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/Accepted password/d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/sudo/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```

Exercise 3: Removing Irrelevant Entries

Without modifying the original file, display the log after:

- Deleting successful login entries.
- Deleting entries containing Connection closed.
- Deleting the first three entries.
- Deleting entries from lines 4 to 7.
- Deleting all entries associated with 192.168.1.15.
- Deleting every second entry starting from the first line.
- Deleting all events that do not contain Failed password.

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
~/Documents/LoS/Lab/Lab5

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/Accepted password/d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/Connection closed/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '1,3d' auth.log
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '4,7d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/192.168.1.15/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '1-2d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/Failed password/d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```

Exercise 4: Classifying Security Events

Replace Accepted password with [SUCCESSFUL LOGIN].

Replace Failed password with [FAILED LOGIN].

Replace Invalid user with [INVALID USER].

Replace authentication failure with [AUTHENTICATION FAILURE].

Replace root with PRIVILEGED_USER.

Replace every occurrence of server with authentication-server.

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
~/Documents/LoS/Lab/Lab5

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/Accepted password/[SUCCESSFUL LOGIN]/g' auth.log

Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/Failed password/[FAILED LOGIN]/g' auth.log

Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/Invalid user/[INVALID USER]/g' auth.log

Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: [INVALID USER] hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/authentication failure/[AUTHENTICATION FAILURE]/g' auth.log

Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: [AUTHENTICATION FAILURE] for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/root/[PRIVILEGED_USER]/g' auth.log

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/root/PRIVILEGED_USER/g' auth.log

Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for PRIVILEGED_USER from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for PRIVILEGED_USER from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for PRIVILEGED_USER from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/server/authentication-server/g' auth.log

Sep 10 09:10:12 authentication-server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 authentication-server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 authentication-server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 authentication-server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 authentication-server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 authentication-server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 authentication-server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 authentication-server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 authentication-server sshd[1235]: Failed password for root from 10.0.0.30
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```

Exercise 5: Masking Sensitive Information

Use sed to anonymise the log before sharing it.

Replace IP address 10.0.0.25 with X.X.X.X.

Replace 172.16.0.8 with SUSPICIOUS_IP.

Replace username admin with USER_A.

Replace username student with USER_B.

Replace all digits with X.

Replace periods in IP addresses with hyphens.

Replace all IP addresses in the sample file with [REDACTED_IP].

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
~/Documents/LoS/Lab/Lab5

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/10.0.0.25/X.X.X.X/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from X.X.X.X
Sep 10 09:12:20 server sshd[1206]: Failed password for root from X.X.X.X
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by X.X.X.Xharis@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/172.16.0.8/SUSPICIOUS_IP/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from SUSPICIOUS_IP
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from SUSPICIOUS_IP
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/admin/USER_A/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for USER_A from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for USER_A from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/student/USER_B/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for USER_B from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user USER_B
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/[0-9]/X/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user USER_B
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/[0-9]/X/g' auth.log
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for admin from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for student from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Invalid user hacker from XXX.XX.X.X
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for admin from XXX.XX.X.X
Sep XX XX:XX:XX server sudo[XXXX]: authentication failure for user student
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for analyst from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
~/Documents/LoS/Lab/Lab5
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```

Part B: Web-Server Log

Use access.log to solve the below questions

Write sed commands to:

Print all requests that returned status code 200.

Print all requests that returned status code 401.

Print all requests that returned status code 403.

Print all requests to /admin/login.

Print all POST requests.

Print requests containing ../.

Print requests containing <script>.

Print requests containing OR.

Print all requests except successful requests.

Print all requests from 10.0.0.25.

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
~/Documents/LoS/Lab/Lab5
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/ 200\r\?$/p' access.log
192.168.1.10 - GET /index.html 200
192.168.1.15 - GET /products.html 200
192.168.1.20 - GET /contact.html 200
192.168.1.25 - GET /about.html 200haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/ 401\r\?$/p' access.log
10.0.0.25 - GET /admin/login 401
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/ 403\r\?$/p' access.log
172.16.0.8 - GET /search?id=1'OR'1='1 403
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.8 - GET /script?input=<script> 403
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/\admin\login/p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/POST/p' access.log
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/\.\.\./p' access.log
10.0.0.30 - GET ../../etc/passwd 403
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/<script>/p' access.log
172.16.0.8 - GET /script?input=<script> 403
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/OR/p' access.log
172.16.0.8 - GET /search?id=1'OR'1='1 403
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/OR/p' access.log
172.16.0.8 - GET /search?id=1'OR'1='1 403
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/^10.0.0.25 /p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```

Exercise 7: Labelling Suspicious Web Requests

Use substitution to label the events:

Replace status code 200 with [ALLOWED].

Replace status code 401 with [UNAUTHORIZED].

Replace status code 403 with [BLOCKED].

Replace /admin/login with /RESTRICTED_RESOURCE.

Replace GET with READ_REQUEST.

Replace POST with SUBMIT_REQUEST.

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
~/Documents/LoS/Lab/Lab5

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/200$/[ALLOWED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html [ALLOWED]haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/401$/[UNAUTHORIZED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/403$/[BLOCKED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/\\admin\\login\\/\\RESTRICTED_RESOURCE/g' access.log
bash: sed: \\admin\\login\\/\\RESTRICTED_RESOURCE/g: No such file or directory
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/\\admin\\login\\/\\RESTRICTED_RESOURCE/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/GET/READ_REQUEST/g' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 's/POST/SUBMIT_REQUEST/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```

Exercise 8: Grouping Security Operations

Use { } to perform multiple commands.

For entries containing status code 401:
 Replace 401 with [UNAUTHORIZED].
 Print the modified line.
 For entries containing status code 403:
 Replace 403 with [BLOCKED].
 Replace the source IP address with [REDACTED_IP].
 Print the modified entry.
 For requests to /admin/login:
 Replace the URL with /PROTECTED_PAGE.
 Replace the source IP with [REDACTED_IP].
 Print the result.

```

haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
~/Documents/LoS/Lab/Lab5
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/ 401$/s/401$/[UNAUTHORIZED]/;p' access.log
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.9 - POST /login [UNAUTHORIZED]
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -En '/403$/s/403$/[BLOCKED]/;s/[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/;p' access.log
[REDACTED_IP] - GET /search?td=1 OR '1'='1 [BLOCKED]
[REDACTED_IP] - GET ../../etc/passwd [BLOCKED]
[REDACTED_IP] - GET /script?input=script= [BLOCKED]
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -En '/\admin\login/s/\admin\login\/PROTECTED_PAGE/g;s/[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/;p;' access.log
[REDACTED_IP] - GET /PROTECTED_PAGE 401
[REDACTED_IP] - POST /PROTECTED_PAGE 401
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$

```

Exercise 9: Adding Incident Labels

Insert the following heading before the first line of auth.log:
SECURITY AUTHENTICATION LOG
 Append the following message after the final entry:
END OF SECURITY LOG
 Insert [WARNING] before every failed login.
 Append Review this event after every invalid-user entry.
 Insert [SUSPICIOUS WEB REQUEST] before every 403 entry.
 Append [REQUIRES INVESTIGATION] after every 401 entry.
 Replace all entries containing Connection closed with:
[CONNECTION TERMINATED]
 Replace all successful web requests with:
[NORMAL WEB REQUEST]

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
~/Documents/LoS/Lab/Lab5

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 'i SECURITY AUTHENTICATION LOG' auth.log
SECURITY AUTHENTICATION LOG
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 'i END OF SECURITY LOG' auth.log
END OF SECURITY LOG
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
END OF SECURITY LOG
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/Failed password/i [WARNING]' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
[WARNING]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
[WARNING]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
[WARNING]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
[WARNING]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
[WARNING]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 'i Invalid user/a Review this event' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Review this event
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/ 403$/i [SUSPICIOUS WEB REQUEST]' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
[SUSPICIOUS WEB REQUEST]
172.16.0.8 - GET /search?id=1'OR'1='1 403
192.168.1.15 - GET /products.html 200
[SUSPICIOUS WEB REQUEST]
10.0.0.30 - GET /../../../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
[SUSPICIOUS WEB REQUEST]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/ 401$/i [REQUIRES INVESTIGATION]' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
[REQUIRES INVESTIGATION]
172.16.0.8 - GET /search?id=1'OR'1='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../../../etc/passwd 403
172.16.0.9 - POST /login 401
[REQUIRES INVESTIGATION]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
[REQUIRES INVESTIGATION]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200haris@ubused 's/.Connection closed.*/[CONNECTION TERMINATED]/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
[CONNECTION TERMINATED]haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```

```

haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/ 200$/c [NORMAL WEB REQUEST]' access.log
[NORMAL WEB REQUEST]
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
[NORMAL WEB REQUEST]
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
[NORMAL WEB REQUEST]
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
[NORMAL WEB REQUEST]
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ 

```

Exercise 10: Separating Log Entries with G

Use G to:

Add a blank line after every authentication event.

Add a blank line after every failed login.

Add a blank line after every blocked web request.

Add a blank line after every second entry starting with the first entry.

Write failed logins to failed_logins.txt.

Write successful logins to successful_logins.txt.

Write invalid-user events to invalid_users.txt.

Write 401 web events to unauthorized_requests.txt.

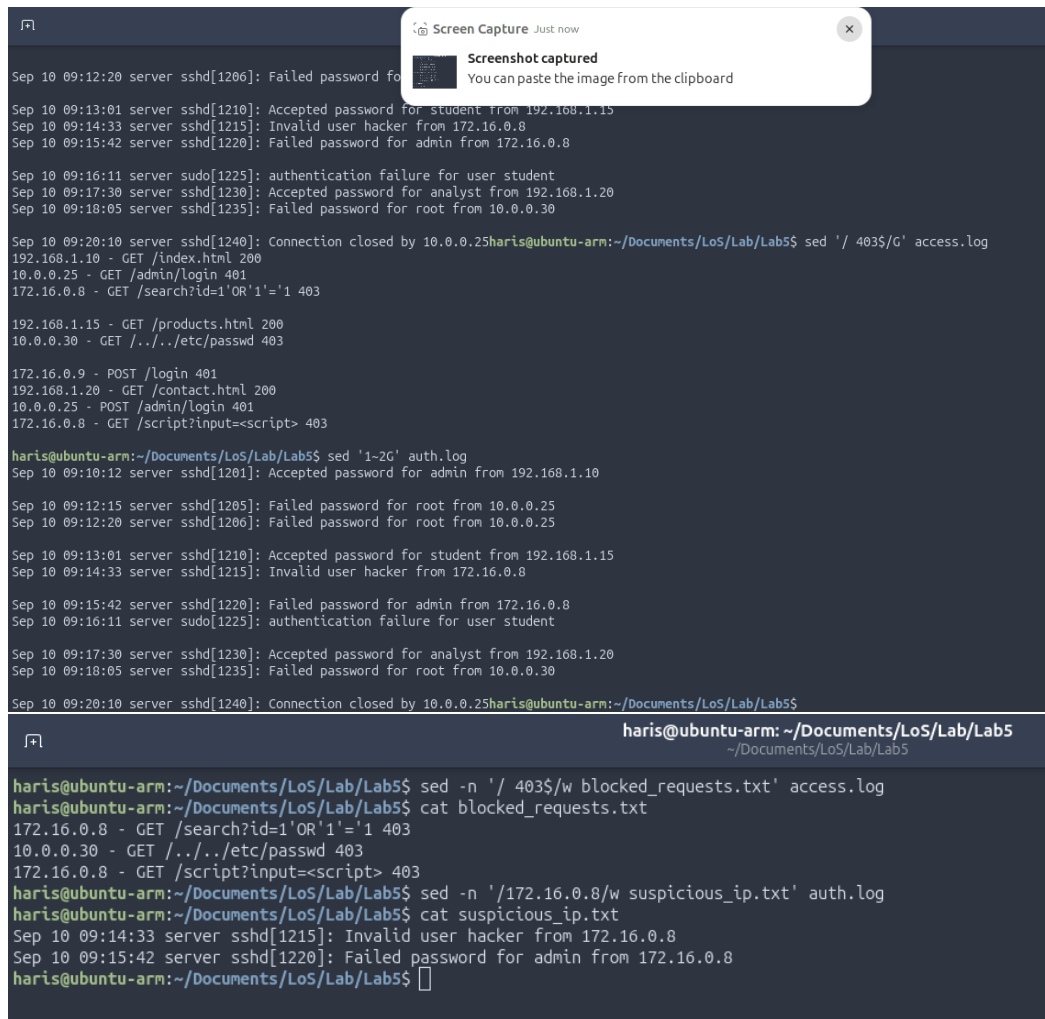
Write 403 web events to blocked_requests.txt.

Write all events related to 172.16.0.8 to suspicious_ip.txt.

```

haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed 'G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/Failed password/G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/ 403$/G' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200

```



```
Sep 10 09:12:20 server sshd[1206]: Failed password for
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '/ 403$/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1='1 403

192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403

172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed '1-2G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student

Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/ 403$/w blocked_requests.txt' access.log
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ cat blocked_requests.txt
172.16.0.8 - GET /search?id=1'OR'1='1 403
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.8 - GET /script?input=<script> 403
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n '/172.16.0.8/w suspicious_ip.txt' auth.log
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ cat suspicious_ip.txt
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```

Label successful and failed logins.

Label 401 and 403 web requests.

Delete successful logins and print failed logins.

Replace root with [PRIVILEGED_USER] and mask all IP addresses.

Replace GET and POST with READ_REQUEST and SUBMIT_REQUEST.

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
~/Documents/LoS/Lab/Lab5

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/g' -e 's/Failed password/[FAILED LOGIN]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -e 's/401$/[UNAUTHORIZED]/' -e 's/403$/[BLOCKED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /search?id=1'OR'1'='1 [BLOCKED]
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd [BLOCKED]
172.16.0.9 - POST /login [UNAUTHORIZED]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /script?input=<script> [BLOCKED]
192.168.1.25 - GET /about.html 200haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -n 's/Accepted password/d;Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -E 's/root/[PRIVILEGED_USER]/g; s/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -e 's/GET/READ_REQUEST/g' -e 's/POST/SUBMIT_REQUEST/g' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST /../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
```

Create security_report.sed containing commands to:

Insert SECURITY INCIDENT REPORT before the first line.

Replace Failed password with [FAILED LOGIN].

Replace Invalid user with [INVALID USER ATTEMPT].

Replace Accepted password with [SUCCESSFUL LOGIN].

Replace all sample IP addresses with [REDACTED_IP].

Append END OF INCIDENT REPORT after the last line.

Execute it using:

sed -f security_report.sed auth.log

Save the output:

sed -f security_report.sed auth.log > incident_report.txt

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
~/Documents/LoS/Lab/Lab5

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ touch security_report.sed
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ cat > security_report.sed <<'EOF'
i SECURITY INCIDENT REPORT
s/Failed password/[FAILED LOGIN]/g
s/Invalid user/[INVALID USER ATTEMPT]/g
s/Accepted password/[SUCCESSFUL LOGIN]/g
s/[0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+/[REDACTED_IP]/g
$ END OF INCIDENT REPORT
EOF
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ cat security_report.sed
i SECURITY INCIDENT REPORT
s/Failed password/[FAILED LOGIN]/g
s/Invalid user/[INVALID USER ATTEMPT]/g
s/Accepted password/[SUCCESSFUL LOGIN]/g
s/[0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+/[REDACTED_IP]/g
$ END OF INCIDENT REPORT
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -f security_report.sed auth.log
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -f security_report.sed auth.log > incident_report.txt
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ cat incident_report.txt
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```

Exercise 14: Modifying a Security Configuration File

Create security.conf:

PermitRootLogin yes

PasswordAuthentication yes

MaxAuthTries 6

FirewallStatus disabled

LogLevel INFO

GuestLogin enabled

First make a backup:

cp security.conf security.conf.backup

Use **sed -i** to:

Change PermitRootLogin yes to PermitRootLogin no.

Change MaxAuthTries 6 to MaxAuthTries 3.

Change FirewallStatus disabled to FirewallStatus enabled.

Change LogLevel INFO to LogLevel VERBOSE.

Change GuestLogin enabled to GuestLogin disabled.

Use the safer backup form:

sed -i.bak 'sed-command' security.conf

```
haris@ubuntu-arm: ~/Documents/LoS/Lab/Lab5
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ cat > security.conf <<EOF
PermitRootLogin yes
PasswordAuthentication yes
MaxAuthTries 6
FirewallStatus disabled
LogLevel INFO
GuestLogin enabled
EOF
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ cp security.conf security.conf.backup
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ cp security.conf security.conf.backup
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -i 's/PermitRootLogin yes/PermitRootLogin no/' security.conf
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -i 's/MaxAuthTries 6/MaxAuthTries 3/' security.conf
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -i 's/FirewallStatus disabled/FirewallStatus enabled/' security.conf
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -i 's/LogLevel INFO/LogLevel VERBOSE/' security.conf
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ sed -i 's/GuestLogin enabled/GuestLogin disabled/' security.conf
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ cat security.conf
PermitRootLogin no
PasswordAuthentication yes
MaxAuthTries 3
FirewallStatus enabled
LogLevel VERBOSE
GuestLogin disabled
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```

This exercise demonstrates configuration editing only; it does not configure the actual system.

Write a Bash script named security_log_analyser.sh.

The script should:

Accept a log filename as a command-line argument.

Check whether the argument is provided.

Check whether the file exists.

Extract failed login attempts.

Extract invalid-user attempts.

Replace sensitive IP addresses with [REDACTED_IP].

Add the heading SECURITY ALERT REPORT.

Add [ALERT] before suspicious entries.

Append END OF REPORT.

Save the result in security_alert_report.txt.

Display the generated report.

```
haris@ubuntu-arm: ~/Documents
~/Documents/LoS/Lab/La

haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$ ./security_log_analyser.sh auth.log
[ALERT] SECURITY ALERT REPORT
[ALERT] Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
[ALERT] Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
END OF REPORT
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab5$
```